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RESEARCH ARTICLE

Vehicle Speed Estimation Using Consecutive Frame Approaches and Deep Image Homography for Image Rectification on Monocular Videos

DEA ANGELIA KAMIL¹, WAHYONO¹, (Member, IEEE), AGUS HARJOKO¹, (Member, IEEE),
AND KANG-HYUN JO², (Senior Member, IEEE)

¹Department of Computer Science and Electronics, Universitas Gadjah Mada, Yogyakarta 55281, Indonesia

²Department of Electrical, Electronic and Computer Engineering, University of Ulsan, Ulsan 44610, South Korea

Corresponding author: Agus Harjoko (aharjoko@ugm.ac.id)

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ABSTRACT In intelligent transportation systems, vehicle speed estimation plays a vital role in traffic monitoring, speed enforcement, and autonomous vehicles. Therefore, the authors proposed a vehicle speed estimation method composed of pipelines: homography transformation using a deep image homography transformation network, vehicle detection by YOLOv8, tracking by ByteTrack, speed estimation in consecutive frames, and regression using Multiple Linear Regression. Homography transformation was utilized to rectify the monocular video view image into a bird's-eye view with a constant pixel-per-meter value. In this scheme, the authors proposed a method for estimating the speed in consecutive frames using statistical and machine learning approaches by comparing several experimental schemes to determine which method is better for estimating vehicle speed, such as vehicle speed estimation results in Root Mean Square Error (RMSE) and Mean Absolute Error (MAE), traffic conditions, vehicle directions, and state-of-the-art studies. A monocular video DeaSpeedDataset was generated to evaluate the proposed system method by comparing the predicted speed value with the ground truth value obtained by the speed gun. This dataset contains 2408 vehicles divided into 20 conditions, four setups, and two locations. Furthermore, the best RMSE was approximately 2.37897 km/h, and the MAE was approximately 1.68977 km/h.

INDEX TERMS Vehicle speed estimation, computer vision, intelligent transportation systems, vehicle detection and tracking, homography transformation, camera calibration.

I. INTRODUCTION

In intelligent transportation systems, vehicle speed estimation is divided into three main domains: traffic monitoring, speed enforcement, and autonomous vehicles [1]. Knowledge of the velocity information of all vehicles allows us to analyze and monitor traffic conditions (traffic flow, density [2], and speed [3]), which can be applied to implementations such as traffic jams, congestion, accident prevention, traffic light time setting, and real-time traffic navigation. Speed estimation

systems in the traffic monitoring domain usually utilize traffic monitoring cameras such as CCTV cameras designed to monitor traffic without any specific platform or setting for estimating vehicle speed. The vehicle speed estimation for speed enforcement is used for road safety measures that cause accidents. The speed limitation depends on the country, and violators of the speed limit are punished. Vision-based speed enforcement sensors were designed to estimate speed with an adequate measuring distance. In autonomous vehicles, calculating the distance and speed of nearby vehicles is crucial for driver assistance and automated driving. Cameras used for autonomous cars are installed on-board the vehicle, such

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as dashboard cameras, and the cameras are combined with range-based sensors such as radar or LiDAR.

One of the most critical parts of vehicle speed estimation systems is the calculation of the relative distance of vehicle displacement. In image perspective theory [4], general parallel lines on the real-world plane are not parallel in the image but instead converge to a finite point. The finite point in the image plane is called the vanishing point [5] and this phenomenon is called the perspective distortion. It is possible to “undo” the perspective distortion by applying an inverse transformation called the homography matrix to the image. Using a homography matrix, the road image can be projected onto a bird’s-eye view image such that the ratio of pixels per meter is constant.

In this study, the authors used a vision sensor based on a monocular camera installed in forehead camera mode captured on a pedestrian bridge. The camera installation settings were necessary to affect the predicted speed results. Some camera settings were used in the experiment to determine the best settings for the speed cameras. Some researchers use variations of differences in sunlight intensity, such as in sunny and rainy weather conditions and morning, afternoon, and evening conditions. In this research, the authors used different vehicle directions to compare which direction had better results and whether different directions affected the speed prediction.

The approach utilized in this study comprises several stages, including image rectification, vehicle detection, vehicle tracking, distance estimation, and speed estimation. To concisely rectify the image from a bird’s eye view, the CNN model was applied to obtain a homography matrix. Calibrating the camera to obtain the actual distance value is a complex task. One of the most popular methods is the use of a homography matrix by formulating calibration and rotation matrix. Generating a calibration matrix using a vanishing point is a common method in computer vision. However, detecting vanishing points using hand-crafted techniques is very complicated, especially if the image contains a significant amount of noise. In [6] and [7], the Deep Image Homography using a CNN successfully transformed a monocular traffic image into a bird’s-eye view image.

Furthermore, in this study, the authors employed YOLOv8 and ByteTrack in the object detection and tracking stages. The reason for using YOLOv8 is used for detection because YOLOv8 has a fast detection capability, high accuracy, and compact model size [8]. Based on [9], YOLOv8 has lower complexity and speedier performance with higher mean average precision (mAP) compared with the other previous versions, YOLOv5, YOLOv6, and YOLOv7 in the COCO validation set. Moreover, in [10], the authors compared Faster R-CNN, SSD, and YOLO. The YOLO family performs better in terms of speed and accuracy than SSD and Faster R-CNN. In terms of selecting ByteTrack for the tracking algorithm, the performance of ByteTrack was better than that of SORT and

DeepSORT. In [11], ByteTrack has a higher Multiple Object Tracking Accuracy (MOTA) of 77.3% than DeepSORT and SORT, with MOTA of 61.4% and 54.7%.

For another comparison, [12] proposed vehicle detection and tracking using a combination of morphological operation motion feature point analysis by the KLT tracker and K-means clustering. The authors in [13] classified and detected vehicles using YOLOv2, and similar to [12], the authors labelled vehicle trajectories by tracking and clustering the corner points using KLT tracker and K-means clustering. In [14], the authors used U-Net and the encoder was initialized by VGG11 weights for building detection and distinguishing the type of damage. In [15], the authors modified YOLOv4 and background subtraction method to apply unsupervised object detection.

Moreover, this study focused on estimating vehicle speed using consecutive frames. In summary, this study makes the following contributions:

- A novel video dataset recorded using a monocular camera was created using an overhead camera installed on a pedestrian bridge. The dataset contains the vehicle speed ground truth obtained manually by a speed gun and recorded using a mobile phone camera.
- The homography transformation network proposed by Yohannes et al. [6] was used for image rectification and pixel-per-meter values were manually applied to estimate the relative distance between travelled vehicles.
- The authors propose a consecutive frame method for estimating vehicle speed. Consecutive or non-consecutive frames can affect the speed. By using consecutive frames for speed estimation, the distance travelled by the vehicle can be estimated automatically, and the vehicle acceleration patterns can be analyzed.
- The vehicle speed results in different directions and under different traffic conditions are compared. The manner in which the camera is installed, whether the vehicle’s direction is toward or away from the camera, and different traffic conditions affect the system and result in accuracy.

The following sections will discuss the whole stages of this study. Section II presents a state-of-the-art vehicle speed estimation study and compares some research in this scope. Section II-A describes all stages of the vehicle speed estimation scheme, starting with how the video dataset is collected, image rectification using the homography transformation network, object detection using YOLOv8, vehicle tracking using ByteTrack, vehicle speed estimation, and regression to improve vehicle speed accuracy using Multiple Linear Regression. Section II-B discusses the experimental results to explain the results of some comparison studies and analyzes which condition is better. Section II-C presents the conclusion of this study, the research’s susceptibility, and possible future work.

TABLE 1. Summary of state of the art on vehicle speed estimation.

Author	Dataset	Hardware	Vehicle Detection			Speed Algorithm	Mean Error (km/h)
			Segmentation	Detection	Tracking		
Nurhad et al. 2013 [22]	Primer	Single	GMM	Hole Filling	Kalman filter	Consecutive Frame	7.63000
Sina et al. 2013 [23]	Primer	Single	Fast Connected-Component Labeling	Blob detection	Euclidean distance	Consecutive Frame	3.32500
Kumar et al. 2014 [24]	Primer	Single	Background subtraction	Corner detection	Feature tracking	Consecutive Frame	2.00000
Dubska et al. [16] 2014	BrnoCS	Single	Background subtraction	Convex hull	Kanade Lucas Tomasi	Consecutive Frame	8.59000
Lin et al. 2015 [28]	Primer	Stereo	Background subtraction	Gaussian Mixture Model	Kalman Filter	Matching Stereo Frames	3.00000
Filipiak et al. 2016 [25]	Primer	Single	N/A	Automatic Number Plate Recognition	N/A	Consecutive Frame	2.00000
Kumar et al. 2018 [18]	AIC18	Single	Mask-RCNN	Mask-RCNN	SORT, DeepSORT	Consecutive Frame	15.35314
Tang et al. 2018 [19]	AIC18	Single	Histogram, Otsu's method	YOLOv2	Multi-camera tracking	Consecutive Frame	6.59235
Huang et al. 2018 [20]	AIC18	Single	N/A	Faster RCNN	Histogram-based tracking	Consecutive Frame	13.85468
Hua et al. 2018 [21]	AIC18	Single	N/A	3D Deformable Model	Kanade Lucas Tomasi	Consecutive Frame	19.48819
Tran et al. 2018 [13]	AIC18	Single	N/A	Faster RCNN Resnet-101	IOU calculation	Non-Consecutive	14.34634
Bouzia et al. 2018 [29]	Primer	Stereo	Background subtraction	Blob detection	SURF	Matching Stereo Frames	2.00000
Sochor et al. 2019 [17]	BrnoCS	Single	Background subtraction	Convex hull	Kanade Lucas Tomasi	Consecutive Frame	1.71000
							15.66000
							1.43000
Trivedi et al. 2021 [26]	Primer	Single	Background subtraction	Interframe difference	Kalman filter	Non-Consecutive	2.87500
Wu et al. 2021 [27]	Primer	Single	N/A	License plate: OCR, YOLO	License plate matching	Consecutive Frame	3.20000
Yohanes et al. 2023 [6]	BrnoCS	Single	N/A	YOLOv4	DeepSORT	Non-Consecutive	2.56125
	AIC18					Non-Consecutive	4.08289
Kamil et al. 2023 [3]	Primer	Single	N/A	YOLOv5	DeepSORT	Non-Consecutive	3.15500

II. RELATED WORK

The number of recent studies on vehicle speed estimation has significantly increased. In this section, the authors discuss several studies on the scope of vehicle speed estimation. Table 1 compares several research papers in terms of their datasets, hardware, segmentation, detection, tracking, speed algorithm method, and mean error values (RMSE or MAE). The following paragraph explains each research approach scheme for estimating vehicle speed.

A. DATASET

In this related work discussion, there are two public datasets that some research has used to evaluate their methods, the BrnoCompSpeed dataset [6], [16], [17] and the NVIDIA AI City Challenge dataset [13], [18], [19], [20], [21]. The BrnoCompSpeed dataset uses a single camera to generate a

traffic-video dataset. The ground truth of the vehicle speed was produced by the LiDAR and GPS modules. Furthermore, the NVIDIA AI City Challenge dataset was generated using a single camera, and ground truth values were obtained using a GPS device. This dataset was created as a challenge in 2018. Other studies have proposed a dataset for evaluating the proposed methods. The ordinary camera used is a single camera [3], [22], [23], [24], [25], [26], [27] while others use stereo cameras [28], [29]. In this study, the authors generated a primer dataset using a single camera, and a speed gun took the ground truth value.

B. VEHICLE DETECTION

Vehicle detection methods are divided into segmentation, detection, and tracking methods. For segmentation, background subtraction [16], [17], [24], [26], [28], [29] and the

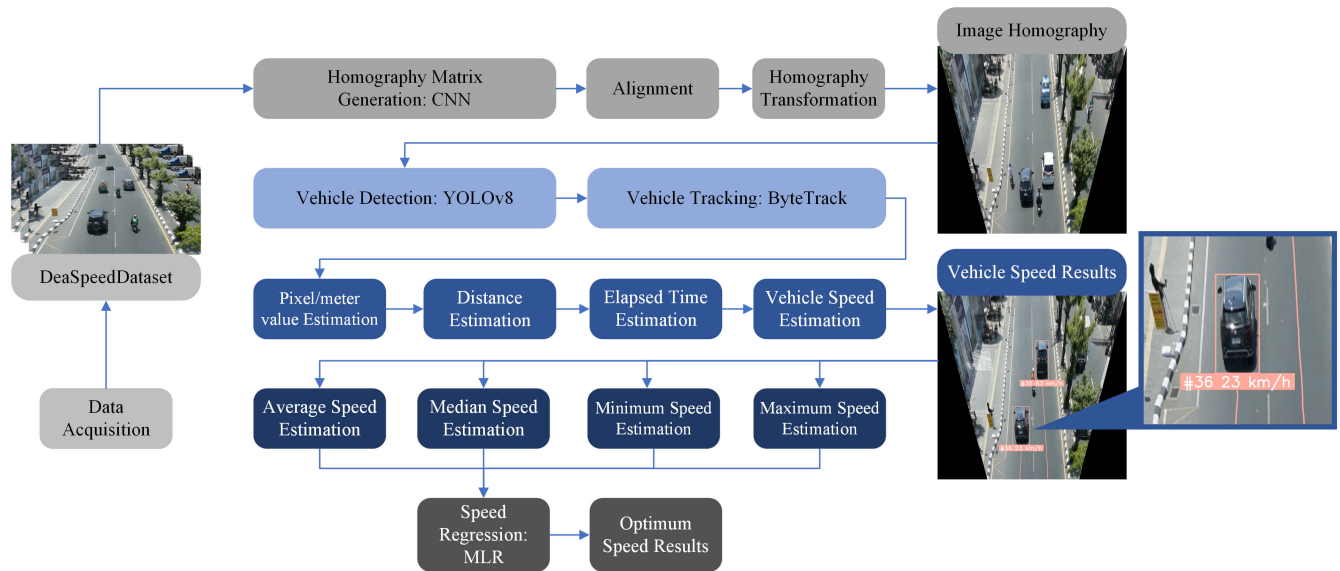


FIGURE 1. Vehicle speed estimation proposed method process.

Gaussian Mixture Model [22], [28] are the common methods used. One learning-based method that can segment an object is the Mask-RCNN [18]. The based method was a common method used for directly detecting vehicles, like the YOLO family: YOLO [27], YOLOv2 [19], YOLOv4 [6], YOLOv5 [3], Mask-RCNN [18], and Faster RCNN [20], [13]. Then, after applying detection, a processed tracking method can be employed. Common methods used in tracking are the Kalman filter [22], [26], [28], Kanade Lucas Tomasi [16], [17], [21], SORT [18], DeepSORT [3], [6], [18], and SURF [29]. The authors adopted YOLOv8 and ByteTrack to detect and track vehicles in this study because they outperformed other methods.

C. SPEED ESTIMATION

Speed estimation methods are divided into two main approaches: consecutive [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [27], and non-consecutive frames [3], [6], [13], [26]. Furthermore, stereo camera-based research has mainly utilized a matching stereo frame approach [28], [29]. The consecutive frame approach uses the distance of the travelled vehicle between frames; this technique can be used as an automatic method for estimating vehicles. The non-consecutive frame approach usually uses a region of interest (RoI) that is measured in length, and the travelled vehicle distance is calculated between the start and end of the RoI. Consecutive methods are difficult to apply because of inaccuracies in detection or tracking; however, they can be used for automatic schemes and to observe vehicle acceleration patterns. In this study, the authors utilized a consecutive frame approach.

III. PROPOSED METHOD

The method in this research was proposed for estimating vehicle speed using a monocular camera. The proposed method

comprises six main steps: data acquisition, image rectification, vehicle detection, speed estimation, statistical, and regression approaches. The entire process is illustrated in Figure 1. The first step in vehicle speed estimation based on the vision sensor scheme is data acquisition which utilizes a single camera to record a traffic view video captured from a pedestrian bridge to capture an overhead camera view. In the evaluation, the ground truth value of the vehicle speed was obtained using a speed gun. In the rectification process, the authors used a CNN model to generate a homography matrix to transform the image into a bird's eye view. YOLOv8 and ByteTrack were used to detect and track vehicles. For vehicle speed estimation, the distance and time of travelled vehicle are required. The distance and time were calculated in consecutive frames. Furthermore, to conclude the several speed results, statistical approaches were used. For improving the accuracy of speed results, regression model was applied. The following sub-chapter discusses the entire process in detail.

A. DATASET

The first essential step in a vehicle speed estimation system is the data acquisition. The authors developed a vehicle speed video dataset with vehicle speed ground-truth values to evaluate the performance of the proposed system. Monocular videos were taken at two different locations, first on the pedestrian bridge of Ambaramarga and second on the pedestrian bridge of the Faculty of Veterinary Medicine, Universitas Gadjah Mada. Both locations are located in Yogyakarta, Indonesia. Each location was divided into two camera setups; thus, the videos were captured in four different camera setups, each containing five videos. The length of each video is approximately 15 minutes. The video data were recorded using a Sonny ZVE-10 Camera with a frame size of 1920×1080 pixel and a frame per second of 30 fps.



FIGURE 2. DeaSpeedDataset, (Upper) first location in Ambaramarga Bridge, (Lower) second location overpass bridge in gate entrance of Faculty of Veterinary Medicine, Universitas Gadjah Mada.



FIGURE 3. Monocular video data acquisition, (Left) data acquisition using single camera and speed radar, (Right) Bushnell radar gun.

To obtaining the ground truth value of vehicle speed, the authors adopted the Bushnell radar, which has an accuracy speed of approximately ± 1.6 kilometer per hour. The images of how the authors took the monocular video dataset and Bushnell radar are shown in Figure 3.

The ground truth values were taken for evaluation in the experimental process and from above the overpass. The speed gun is only shot on four-wheeled vehicles, including cars, buses, and trucks; the other vehicles (e.g. motorcycles) are ignored owing to the limitation of human power to catch all

vehicle speeds that pass the road and not all cars shown in the video have ground truth value. The total number of vehicles included in the experiment was 2408. The camera settings are designed to capture different vehicle directions, and it takes in tilted-down mode, so it helps the homography transformation network rectify the image successfully.

Images of the video dataset are shown in Figure 2. The two locations on this dataset had their own characteristics. The first location on the Ambaramarga Bridge has an average vehicle speed with a higher value because it is located on the main road with a higher traffic flow rate. The lowest value of the recognized vehicle was approximately 16 km/h, and the highest was approximately 67 km/h. Meanwhile, in the second location of the Faculty of Veterinary Medicine, Universitas Gadjah Mada, the passing vehicles shown in the video had lower speeds because the recorded spot was located at the campus entrance gate. The lowest vehicle speed value at this location was approximately 16 km/h, and the highest was approximately 46 km/h.

However, there are some challenges in dealing with monocular videos. A monocular image is a scene from a camera that models the perspective projection from 3D space into an image that allows two adjacent objects that should be parallel in the real world to appear closer. The point where the objects are not far apart is called the vanishing point [4], [5]. Based on this characteristic, a monocular image has different distance scales or pixel-per-meter values, even in one image plane. Moreover, unlike stereo cameras, which have an exact

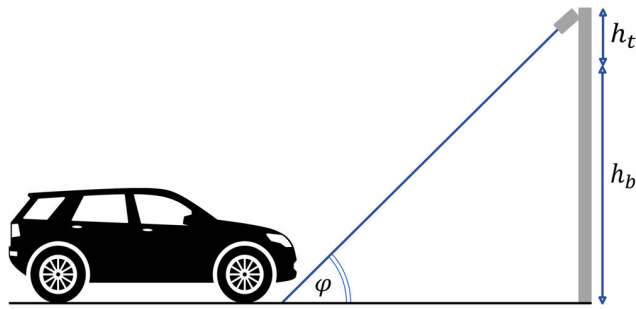


FIGURE 4. Camera installation.

distance value between the cameras, some monocular image datasets do not provide distance-scale information. Therefore, it is difficult to estimate the scale factor in monocular images [30]. To deal with these challenges, the authors transformed the monocular videos into a bird's-eye view to rectify the road view into a parallel form so that the scale factor has a constant value. Then, to deal with scale ambiguity, the authors provide scale factor information in Section III-F, which was calculated using laser meter based on the length of the road marking lines.

An illustration of the camera installation is shown in Figure 4, where h_t is the height of the camera tripod, h_b is the height of the bridge, and φ is the camera angle. The camera is installed above the bridge, Ambaramarga Bridge has a bridge height h_b of 6.3 m, and the Faculty of Veterinary Medicine Bridge has a bridge height h_b of 5.7 m, with a tripod height h_t of about 1.5 – 1.7 m in each installation. The camera angle φ was also measured in each installation within a range of $2^\circ \leq \varphi \leq 14^\circ$. The height and angle of the camera were measured using a laser meter. The vehicle speed value was in the range of 16 – 67 km/h.

B. PRELIMINARIES

In [4], homography or projective transformation is defined as an invertible mapping of points and lines from points in a 2D projective plane to itself, such as three collinear points. In this study, the homography matrix was used to rectify the monocular traffic image into a geometrically correct bird's eye view. The well-known bird's-eye view is an overhead or orthographic view in which the lines in the image appear to be orthogonal or parallel to each other [4], [7]. For a better explanation, please refer to Figure 5. The original image P is in the monocular image form. Using a homography matrix H , the monocular form can be transformed into the target image P' or a rectified view image (in bird's-eye view) by removing the perspective distortion. This relation can be formulated using Equations (1) and (2):

$$P' = H \cdot P, \quad (1)$$

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = \begin{bmatrix} h_1 & h_2 & h_3 \\ h_4 & h_5 & h_6 \\ h_7 & h_8 & h_9 \end{bmatrix} \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}, \quad (2)$$

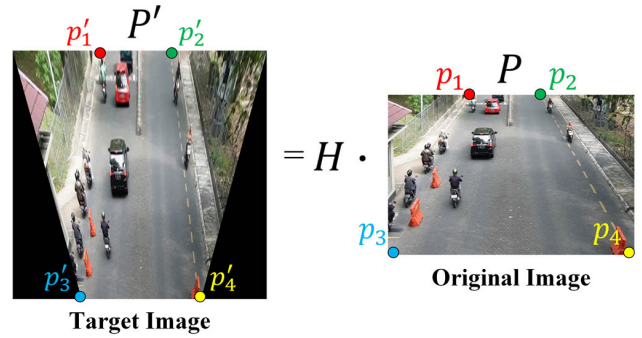


FIGURE 5. Homography application for image rectification.

where H is a 3×3 . Meanwhile, each pair of points is defined as $p'_i = [x'_i \ y'_i \ 1]^T$ and $p_i = [x_i \ y_i \ 1]^T$.

C. HOMOGRAPHY PREDICTION

The authors refer to the relationship between the two images in Figure 5, P and P' with the same camera center [4], [7]. The two images P and P' are interconnected by homography. Where the camera is rotated about its center with no transformation in the internal parameters, algebraically, with x , x' being the image pixels of a scene coordinates X before and after the pure rotation, the relation can be written as Equations (3) and (4):

$$x = K[I \ 0]X, \quad (3)$$

$$x' = K[R \ 0]X = KRK^{-1}K[I \ 0]X = KRK^{-1}x, \quad (4)$$

such that $x' = Hx$ where $H = KRK^{-1}$. The internal calibration matrix K is then expressed by Equation (5):

$$K = \begin{bmatrix} f & 0 & w/2 \\ 0 & f & h/2 \\ 0 & 0 & 1 \end{bmatrix}, \quad (5)$$

where f is the camera focal length, w and h are the width and height of the image, respectively.

The matrix K can be calculated by determining the focal length f of the camera. Based on the explanation in [4], the focal length can be determined using the relationship in Equation (6):

$$l = \omega v, \quad (6)$$

where, l is the horizon line, ω is called the absolute conic image, and v is the vertical vanishing point. The image of the absolute conic is uninfluenced by camera rotation and can be written as $\omega = (KK^T)^{-1}$. The camera is assumed to have zero skew for determining the focal length, the pixels are assumed to be square (or equivalently, their aspect ratio is known), and the image center is assumed to be the principal point. The image of the absolute conic ω can be described as a diagonal matrix $\text{diag}(1/f^2, 1/f^2, 1)$ with only one degree of freedom. Therefore, using Equation (6), the focal length formula can

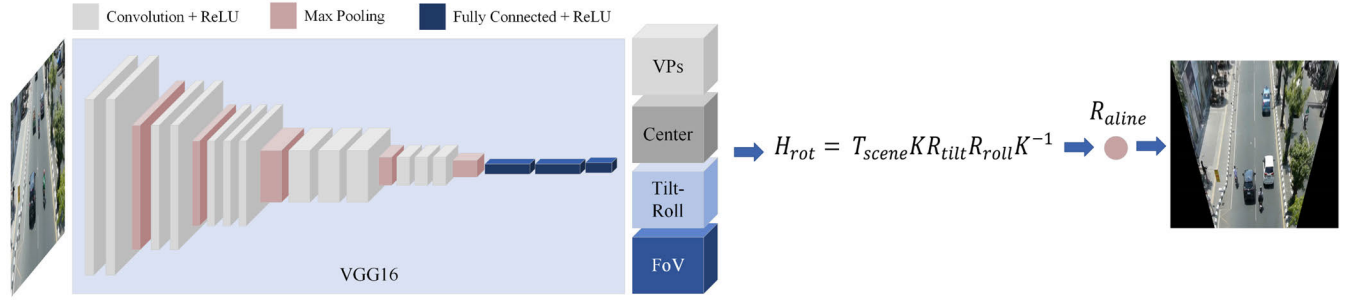


FIGURE 6. Architecture of homography transformation network.

be written as Equation (7):

$$f = \sqrt{\frac{v}{l}}. \quad (7)$$

In Equation (4), the rotation matrix R of the rotations about different axes can be arranged. First, the camera was rotated about its principal axes to rectify the camera roll. Second, the camera was rotated about the x-axis to achieve an overhead view of the scene. Next, the sequence of projective transformation is described.

First, a rotation was applied in the principal axis to remove the camera roll such that the camera x-axis can parallel the world's x-axis. The roll of the camera can be computed from the given horizon line $ax + by + c = 0$, the roll of camera can be computed. The camera roll θ_z can be expressed as in Equation (8):

$$\theta_z = \tan^{-1} \left(\frac{-a}{b} \right). \quad (8)$$

The rotation matrix R_{roll} for rotation in the principal axis was calculated using the camera roll θ_z . The next step was to remove the camera tilt to rotate the images about the x-axis of the camera. The camera tilt θ_x can be determined using focal length and either the horizon line or vanishing point. The tilt θ_x can be calculated using Equation (9):

$$\theta_x = \frac{\pi}{2} - \tan^{-1} \left(\frac{\|v_z\|}{f} \right), \quad (9)$$

where f is the camera focal length and $\|v_z\|$ is the perpendicular distance between the vertical vanishing point and the principal point. Therefore, the homography matrix H_{rot} can be written as shown in Equation (10):

$$H_{rot} = KR_{tilt}R_{roll}K^{-1}, \quad (10)$$

where R_{tilt} is the rotation matrix on the x-axis and is used to restore the camera tilt.

Furthermore, once the camera rotation effect is complete, the camera must be translated. Image translation can be achieved using H_{rot} in Equation (10). The translation matrix T_{scene} maps the corners returned to the corners from the previous step image. The last optional step is calculating R_{aline} , which involves aligning the primary direction in the image axes in the Cartesian coordinate system by rotating the

final image by an angle θ_{aline} . The final step is discussed in the following sub-section. The last step is to compose all of the above transformations into the final homography matrix given in Equation (11):

$$H_{rot} = R_{aline}T_{scene}KR_{tilt}R_{roll}K^{-1}. \quad (11)$$

D. DEEP HOMOGRAPHY

The deep homography transformation network method was adopted from the research proposed by Yohannes et al. [6]. The purpose of applying a homography transformation network is to transform a point-of-view image from a tilted-down camera with an infinite vanishing point into a bird's eye-view image. The reason for warping the image into a bird's-eye view was that the pixel-per-meter value of the image was constant. In other words, homography transformation was used to calibrate the images.

According to Sochor et al. [7], there are four steps before obtaining the variable to generate a homography matrix, as discussed in the previous sub-section. In [7], the details of ground truth generation were discussed. The horizon line and vanishing point are the main ideas of Carla-VP ground truth generation. CARLA is a simulator built over Unreal Engine 4, which can generate photo-realistic images with varying fields of view, roll, tilt, and height of the camera under various environmental conditions.

The focal length can be generated from the field of view using Equation (12):

$$\tan \left(\frac{\gamma}{2} \right) = \frac{w}{f}, \quad (12)$$

where γ is the field of view and w is the image width. The founding of focal length implies that the calibration matrix K is also known. Furthermore, the varying tilt, roll, or yaw in the CARLA can give the value of the camera rotation matrix $R = [r_1 \ r_2 \ r_3]$ by composing it as an individual rotation matrix. Using the known K and R , the ground truth for the orthogonal vanishing points can be generated.

Taking into consideration a point at infinity in the z direction, z_∞ , represented by $[0 \ 0 \ 1 \ 0]^T$ in homogenous coordinates. Subsequently, the camera projection in Equation (13)



FIGURE 7. Deep Homography result on deaspeeddataset.

is used:

$$v_z = K[R \mid t]z_\infty = K \begin{bmatrix} r_1 & r_2 & r_3 & t \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix},$$

$$v_z = Kr_3. \quad (13)$$

This formulation can be applied in the same manner to solve the orthogonal horizontal vanishing points given by $v_x = Kr_1$ and $v_y = Kr_2$, and the horizon line can be written as $l = v_x \times v_y$.

Figure 6 shows the working of the architecture of the homography transformation network. The homography transformation network contained VGG-16 as the backbone and generated a 3×3 homography matrix. To train the VGG-16 model, the authors used the Carla-VP2 dataset, which can be accessed at (<https://carla.org/>). The dataset consists of the ground truth values of the vanishing points (v), center (x_s, y_s), tilt-roll (θ_x, θ_z), and field of view (γ_w, γ_h). After the training process, the vanishing points, center, tilt-roll, and focal of view values were utilized to calculate the homography matrix (H_{rot}) using Equation (14) without the R_{aline} :

$$H_{rot} = T_{scene}KR_{tilt}R_{roll}K^{-1}. \quad (14)$$

After transforming the monocular image into a bird's-eye view image using a homography transformation network, the image is randomly skewed so that the rotation function is applied R_{aline} to align the image so that the frame is as straight as possible. A probabilistic Hough transform was adopted to align the frame and discover the line segments. Then, the gradient (m) between two coordinates (x_1, y_1, x_2, y_2) of a line is shown in Equation (15), and the angle ($\phi_{x,y}$) in

Equation (16):

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad (15)$$

$$\phi_{x,y} = \tan^{-1} |(m_2 - m_1) / (1 + (m_1 \times m_2))| \quad (16)$$

Using Deep Image Homography, the authors transformed the traffic video frames of the DeaSpeedDataset within two different camera directions in two locations. After aligning the image using the Hough transform, the entire image-transformation step is complete. The results of the bird's-eye image from deep homography on the DeaSpeed-Dataset traffic video are shown in Figure 7.

E. VEHICLE DETECTION AND TRACKING

The following steps were used for vehicle detection and tracking. In this research, the authors utilized YOLOv8 [31] for detection and ByteTrack [32] to track the vehicle. The authors also implemented the process of detecting and tracking the vehicle using Supervision [33] by the Roboflow library. Vehicle detection and tracking are crucial parts of vehicle speed estimation. The results of these two processes affect the estimated speed.

YOLOv8 had been released by Ultralytics in 2023. YOLOv8 provides five scaled versions: YOLOv8n (nano), YOLOv8s (small), YOLOv8m (medium), YOLOv8l (large), and YOLOv8x (extra-large). The YOLOv8 architecture utilizes a modified CSPDarknet53 backbone from the YOLOv5. Instead of the CSPLayer, YOLOv8 uses a C2f module (cross-stage partial bottleneck with two convolutions), combining high-level features with contextual information to improve detection accuracy. A spatial pyramid pooling fast (SPPF) layer accelerates the computation by pooling the features into a fixed-size map. Each convolution had batch normalization and SiLU activation. In this research, the authors used pre-trained YOLOv8x using the MS COCO dataset test-dev 2017, which achieved an AP of 53.9% compared with YOLOv5 of 50.7% [9].

After applying YOLOv8x to detect vehicles, multi-object tracking using ByteTrack is the next step. In contrast to previous multi-object tracking methods (SORT [34] and DeepSORT [35]), which only keep the high score detection boxes, ByteTrack keeps almost every detection box and separates them into high score and low scores in accordance with the score threshold. After the separation process, a Kalman filter is adopted to predict the new locations in the current frame for each track. The high-score detection boxes are associated with all the tracks using the metrics similarity by IoU or Re-ID. Then, the Hungarian Algorithm is adopted to complete the matching based on similarity [32]. Based on the comparison in [11], ByteTrack consistently outperformed SORT and DeepSORT across most evaluation metrics.

F. VEHICLE SPEED ESTIMATION

To estimate the vehicle speed, the authors used a physical formula containing the ratio between the distance of the traveled vehicle and elapsed time within kilometer-per-hour (km/h).

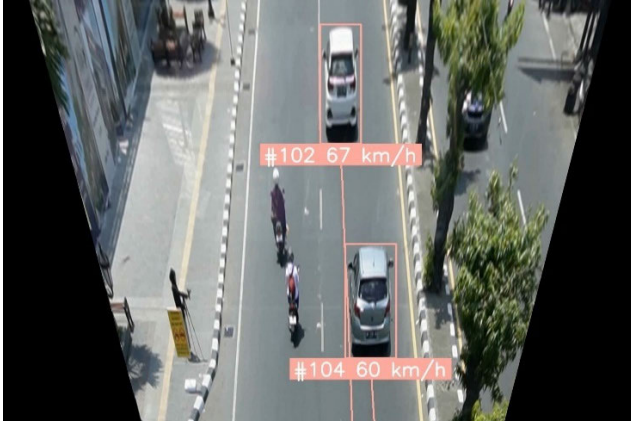


FIGURE 8. Bounding boxes point tracking for estimating vehicle speed.

In this study, the authors proposed a consecutive frame method using four statistical formulas: average, median, minimum, and maximum. To compute the vehicle speed value, the system equation is given by Equations (17), (18), and (19):

$$T_i = \frac{\|t_i\|}{fps}, \quad (17)$$

$$\lambda = \frac{pixel}{meter}, \quad (18)$$

$$V = \frac{|D_{i-\tau} - D_i|}{\lambda |T_{i-\tau} - T_i|} \times (3.6), \quad (19)$$

where $i = 1 \dots N$ is the index for sequence frames, τ is a constant value for controlling the distance between frames, $\|t_i\|$ is the number of frames from the previous when the vehicle detected by the detector to the current frame i , T_i is the ratio of the number of frames per frame per second (fps). Furthermore, D_i is the coordinate point of the vehicle in the image domain. The ratio of pixels per meter is denoted as λ , and the ratio between pixels per meter maps the relation between the image and real-world domain.

In computer vision applications, research involving distance estimation and mapping of the relation between the image domain and real-world domain is critical. In this study, because the video frames were rectified into a bird's-eye view using deep image homography, the pixel-per-meter ratio is a constant value for the entire road in each video location. The pixel-per-meter ratio λ values differed for each video because the pixel-per-meter value changed depending on the camera setting.

In this study, this constant value was manually calculated using the length of the road-marking lines. At the first location, Ambaramarga Bridge, the road-marking lines were 2.5 meters long. Furthermore, in the second location, the Faculty of Veterinary Medicine Bridge, the length of the road-marking lines is 1 meter. However, specifically for the third video setup, because the marking lines are not in the video, the authors assume that the set of black and white middle separator roads is the same with one line marking approximately 1 meter in length.

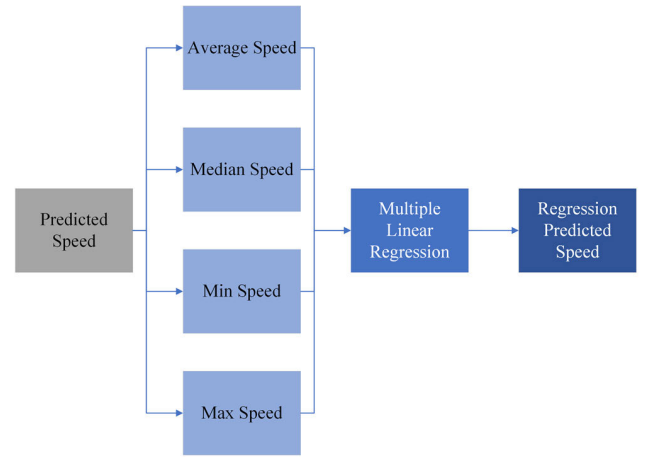


FIGURE 9. Multiple linear regression scheme for predict vehicle speed.

In this research, the authors use a bounding box point as the vehicle coordinate D_i , located in the middle of the bounding box bottom; an illustration of bounding box point tracking is shown in Figure 8. The coordinate point was placed at the bottom of the bounding box to represent the point closest to the ground. Thus, the distance of the traveled vehicle computed from the coordinate point is geometrically correct because the pixel per meter value is taken from the length of the road marking lines, and the homography matrix is focused on rectifying the road image from the bird's-eye point of view.

Subsequently, for vehicle speed V , the authors utilized four different statistical algorithms to compute several vehicle speed values in consecutive frames, which refer to Equations (20), (21), (22), and (23) where the average V_{ave} , median V_{med} , minimum V_{min} , and maximum V_{max} :

$$V_{ave} = \text{average}_{i=1 \dots N-\tau} \left(\frac{|D_{i-\tau} - D_i|}{\lambda |T_{i-\tau} - T_i|} \times (3.6) \right), \quad (20)$$

$$V_{med} = \text{median}_{i=1 \dots N-\tau} \left(\frac{|D_{i-\tau} - D_i|}{\lambda |T_{i-\tau} - T_i|} \times (3.6) \right), \quad (21)$$

$$V_{min} = \text{min}_{i=1 \dots N-\tau} \left(\frac{|D_{i-\tau} - D_i|}{\lambda |T_{i-\tau} - T_i|} \times (3.6) \right), \quad (22)$$

$$V_{max} = \text{max}_{i=1 \dots N-\tau} \left(\frac{|D_{i-\tau} - D_i|}{\lambda |T_{i-\tau} - T_i|} \times (3.6) \right). \quad (23)$$

The objective of using four different statistical algorithms instead of one was to determine which algorithm had the minimum error compared to the ground truth value. Vehicle speed experiments using four different algorithms are discussed in Section IV.

G. VEHICLE SPEED ESTIMATION USING MULTIPLE LINEAR REGRESSION

To improve the estimation speed, the authors adopted a Machine Learning method, Multiple Linear Regression [36], to predict more accurate vehicle speed values through consecutive frame approaches. The Multiple Linear Regression method was chosen because, in the comparison study by

Rodríguez-Rangel et al. [37], Linear Regression had the best performance accuracy compared with the other machine learning methods. In this study, four statistical values of the estimated vehicle speed in consecutive frames were used as multiple features to train the regression model.

The Multiple Linear Regression model is an expanded version of the Linear Regression method, which summarizes the relationship between numerous predictor variables to predict a single response variable. The general form of the Multiple Linear Regression equation is written by the following Equation (24):

$$\hat{Y}_i = b_0 + b_1X_{i1} + b_2X_{i2} + \dots + b_jX_{ij} + \dots + b_pX_{ip}, \quad (24)$$

where $X_{i1}, X_{i2}, \dots, X_{ip}$ are the scores of case i on the $b_1, b_2, \dots, b_j, \dots, b_p$ which are partial regression coefficients or regression weights where, b_0 is the regression intercept.

In the consecutive frame method for estimating the vehicle, predicting the accurate speed value is difficult to execute because the consecutive frame is generated throughout the vehicle moving along the road visible in the video, whereas the ground truth value is generated only once in a specific part of the road. Therefore, statistical methods use a regression model to improve the accuracy of the predicted vehicle speeds. The four statistical vehicle speed methods V_{ave} , V_{med} , V_{min} , and V_{max} are used as features or independent variables to predict the dependent variable or predicted vehicle speed by regression V_{reg} . The authors compared 70% and 30% of the training and testing data, respectively. The multiple linear regression process for predicting vehicle speed in consecutive frames is illustrated in Figure 9.

H. EVALUATION METRICS

To evaluate the proposed vehicle speed estimation system, the authors used Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) in kilometers per hour (km/h). These two evaluation metrics are commonly used to evaluate vehicle speed estimations. The lower the value of the RMSE or MAE, the better the speed value prediction. Otherwise, the higher the value of RMSE or MAE, the worse the speed value prediction. The equations for RMSE and MAE are shown in Equations (25) and (26) respectively:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{r}_i - r_i)^2}, \quad (25)$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{r}_i - r_i|, \quad (26)$$

where, $\hat{r}_i$ is the predicted value, r_i is the actual value in the dataset for $i = 1, 2, \dots, n$ vehicles, and n is the number of instances.

IV. EXPERIMENTAL RESULTS

In this section, the experimental studies are discussed. The first concerns detailed information on the DeaSpeed-Dataset. The second is the experimental result of vehicle

speed estimation, with data obtained using the RMSE and MAE. Furthermore, a comparative study of traffic conditions and vehicle direction is examined. Then, state-of-the-art research is compared with the proposed method. The system was run on a computer using an Intel®Core™i9-10900x CPU 3.70GHz with RAM 128 GB, GPU NVIDIA GeForce RTX 2080 Ti with VRAM 12 GB, and Google Colaboratory using GPU Tesla T4 with 16 GB VRAM. The sample results of this research were published in the following link: <https://drive.google.com/drive/folders/1UbDXVRvyBdXh7oO6vF-992LWK-fo7s8L>.

A. DEASPEEDDATASET DETAIL INFORMATION

Table 2 presents the details of the DeaSpeedDataset. The table description of the dataset contains the number of ground truths, minimum and maximum vehicle speed values, number of video frames, and the duration of the video. To make it easier to understand, in this experimental study, the first location is called *Amplaz*, and the second location is called *FKH*. Both locations were divided into two setups.

Generally, the four setup videos had different maximum vehicle speed values because the roads had different traffic conditions. The first location had a higher traffic flow than the second location because the first location was on the main road, whereas the second location was on the university road near the entrance gate. For the first locations, the first and second setups had slightly different traffic conditions. The first setups had lower traffic flow than the second setup. Similar to the first location, in the second location, the first setups have a lower traffic flow than the second setups. Furthermore, the duration of the video is almost the same, approximately 15 minutes, as well as the number of frames, in addition to *FKH01a*, which is the most extended duration of approximately 17 minutes, and *FKH02e*, which is the shortest duration of approximately 13 minutes.

B. VEHICLE SPEED ESTIMATION RESULTS

This experimental study discusses the speed estimation results by comparing RMSE and MAE metrics. The RMSE and MAE metrics were used to measure the errors in the predicted speed compared with the ground truth value generated by the speed gun. The calculation of the RMSE and MAE can be seen in Equations (25) and (26). The authors used a kilometer-per-hour (km/h) in this comparison. Table 3 shows the RMSE, and Table 4 shows the MAE. Both tables contain the average, median, minimum, maximum, and Multiple Linear Regression (MLP) speed error values. The predicted vehicle speeds for each setup location are shown in Figure 10.

In Tables 3 and 4, the authors discuss the RMSE and MAE results of the vehicle speed of the average, median, minimum, maximum, and MLP. In addition to the MLP speeds, vehicle speed estimation uses the statistical relations of the vehicle speed in consecutive frames. Along the road, the speed of a vehicle experiences a change in speed called acceleration or deceleration. Speed changes can be determined by estimating the vehicle speed using the consecutive frame approach.

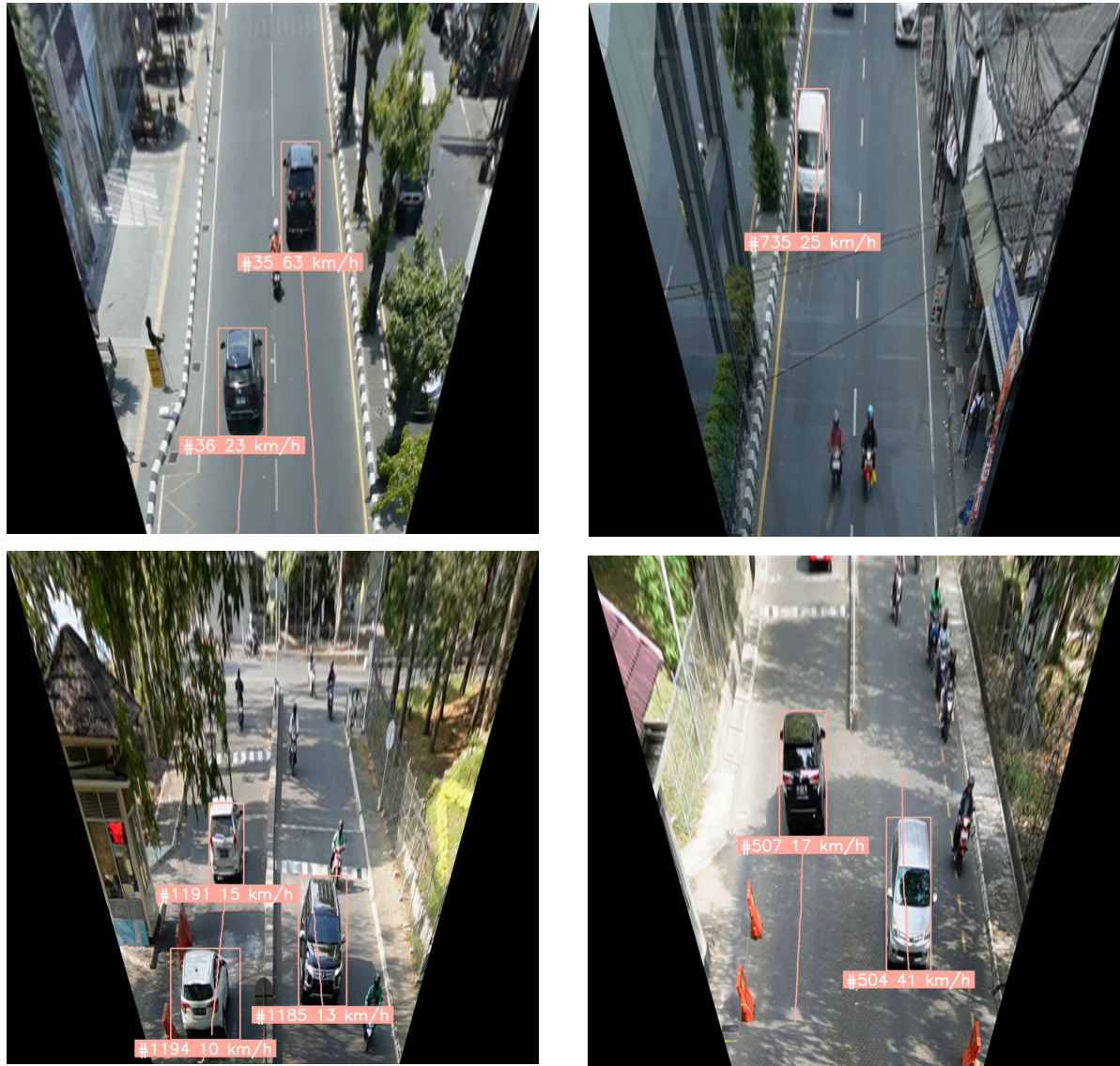


FIGURE 10. Vehicle speed estimation result on bird's-eye view image.

However, it is difficult to determine the speed of a vehicle at one point using only statistical methods because the ground truth value is only taken from one particular location.

It can be observed in Tables 3 and 4 that if the RMSE and MAE values of the minimum speed are small, then the maximum speed error value is high. This relationship occurs because of the speed alteration. If the minimum speed value is the speed value closest to the ground truth, then the maximum speed value is the value farthest from the ground truth, and vice versa. Furthermore, the average and median speed can best represent consecutive frame speed if the moving vehicles do not experience significant speed alteration. If the average speed is the mean value of the speed in successive frames, the median speed represents the intermediate value between the quickest and slowest speeds.

The RMSE results are discussed in Table 3. For the statistical approach of speed estimation including average, median, minimum, and maximum speed, the best-predicted speed value is reached by the maximum speed in *FKH01b* setup at approximately 3.66275 km/h. However, the largest RMSE from the maximum speed approach, *Amplaz02a*, was approximately 32.20589 km/h. Otherwise, the best result in the minimum speed is the highest. However, the worst value of the minimum speed is not higher than that of the maximum speed. This proves that the minimum and maximum speeds cannot be the optimal representation of the speed value of all consecutive frames even though, in this experiment, the maximum speed has the optimal predicted value compared with the statistical method.

Furthermore, as shown in Table 3, the average and median speeds have slightly different RMSE values. The nearest

TABLE 2. Deaspeeddataset detail information.

Locations	Ground Truth	Min (km/h)	Max (km/h)	Frames	Duration
Amplaz01a	141	18	50	27101	15:03
Amplaz01b	175	16	48	27476	15:15
Amplaz01c	196	18	53	27176	15:05
Amplaz01d	217	17	49	27266	15:08
Amplaz01e	231	17	55	27176	15:05
Amplaz02a	155	19	58	27656	15:21
Amplaz02b	130	29	67	27611	15:20
Amplaz02c	122	24	58	27806	15:26
Amplaz02d	110	26	62	27461	15:15
Amplaz02e	172	25	61	27491	15:16
FKH01a	78	17	31	30690	17:03
FKH01b	89	16	37	27417	15:13
FKH01c	83	16	33	27747	15:24
FKH01d	90	16	31	27477	15:15
FKH01e	73	16	46	27267	15:08
FKH02a	39	17	37	27162	15:05
FKH02b	62	17	36	28228	15:40
FKH02c	87	17	46	27462	15:15
FKH02d	73	17	33	27537	15:17
FKH02e	85	16	32	24324	13:30

difference was only approximately 0.02769 km/h, and the largest difference was only 1.79974 km/h. The maximum differentiation was 2 km/h. However, the RMSE of the average and median speeds were, at best, 6.18919 km/h and 6.26522 km/h, respectively. From the statistical approach results, the best result was reached by the maximum speed method (3.66275 km/h), followed by the average speed (6.18919 km/h), median speed (6.26522 km/h), and minimum speed (3.66275 km/h). The authors aimed to improve the accuracy of the predicted results so that a regression algorithm could be applied. It is shown that MLR practically dominated the predicted speed for all setups except one. The MLR method consistently improved the predicted speed from *Amplaz01a* until *FKH02e* except for *FKH01e* with only a 0.09756 km/h difference between MLR and the best result from the statistical method at this setup (maximum speed). *FKH01c* achieved the best result in MLR at approximately 2.37897 km/h, which was also the best result for the entire setup.

Table 4 shows the MAE results of vehicle speed estimation using the DeaSpeedDataset. In general, the trend of the MAE results was similar to that of the RMSE, in which the MAE had a smaller error value than the RMSE result because of the difference in the formula used between the RMSE and MAE evaluation metrics. The MAE value tends to always be smaller than the RMSE. Unlike RMSE, all MLR results were better than those of the statistical methods except *FKH01b*, whereas in RMSE, it was *FKH01e*. However, the difference between the MLR results and the highest value of the statistical method (maximum speed) was

TABLE 3. RMSE of vehicle speed estimation on DeaSpeedDataset.

Locations	Average	Median	Min	Max	MLR
Amplaz01a	19.17294	18.97974	12.84772	30.47043	4.71367
Amplaz01b	15.27217	15.78997	13.90733	25.57320	6.13980
Amplaz01c	9.13658	9.10889	12.73153	17.40118	6.53587
Amplaz01d	7.37288	7.58250	14.03600	14.91744	4.91177
Amplaz01e	8.62316	9.20474	16.38102	12.05992	6.37117
Amplaz02a	19.84139	20.52457	16.10189	32.20589	7.84196
Amplaz02b	19.47699	20.79257	15.20374	28.99748	6.56763
Amplaz02c	19.57401	19.67476	14.82798	31.77211	5.53913
Amplaz02d	18.57101	20.16015	14.52177	30.78828	6.30081
Amplaz02e	20.81223	21.83923	14.89830	32.17639	7.18589
FKH01a	14.11318	15.91292	21.62264	4.55170	2.82246
FKH01b	11.67257	13.20187	18.11139	3.66275	3.36895
FKH01c	13.04168	14.41207	20.10604	3.72229	2.37897
FKH01d	8.02405	8.75389	15.01777	5.86610	2.47046
FKH01e	10.60142	11.69958	17.41870	5.47472	5.57228
FKH02a	8.25628	8.91736	15.16998	12.44887	4.77025
FKH02b	8.99723	10.27152	16.23964	12.97081	3.03281
FKH02c	8.58535	9.42347	14.20134	16.73251	4.46648
FKH02d	6.93222	7.31694	13.07617	11.47242	3.02802
FKH02e	6.18919	6.26522	7.40985	14.28615	3.67148
Best Result	6.18919	6.26522	7.40985	3.66275	2.37897

not significantly different, only approximately 0.35948 km/h. The order of the best statistical method is the same as that of the previous evaluation metrics: maximum speed has the best result with 2.08989 km/h, followed by an average speed of 4.66708 km/h, a median speed of 4.75294 km/h, and a minimum speed of approximately 6.29412 km/h. However, the Multiple Linear Regression method yielded the best result, with only a MAE value of 1.68977 km/h.

C. COMPARISON ON TRAFFIC CONDITIONS

Table 2 lists the information on the traffic conditions of each location. From these data, the relationship between different traffic conditions and the predicted speed results can be analyzed. The traffic conditions in the *Amplaz01* and *Amplaz02* setups had higher traffic flows than those in *FKH01* and *FKH02* setups, respectively. Moreover, the first location had a higher vehicle speed range than the second location. The *Amplaz01* setup had a lower speed range than that of *Amplaz02*. Similarly, the *FKH01* setup had a speed range that was less than that of *Amplaz02*.

The level of traffic flow and speed range affect the speed prediction results. It can be observed in Tables 3 and 4 that the higher the speed range at a location, the greater the RMSE and MAE values. In *Amplaz01*, the smallest RMSE was 4.71367, and the highest was 6.37117. In *Amplaz02*, which has a higher speed range, the smallest RMSE is 5.53913, and the largest is 7.84196. Furthermore, in the second location with a lower speed range, *FKH01*, the smallest RMSE was 2.37897, and the largest was 5.47472. For *FKH02*, the lowest RMSE was 3.02802 and the greatest was 4.77025.

TABLE 4. MAE of vehicle speed estimation on DeaSpeedDataset.

Locations	Average	Median	Min	Max	MLR
Amplaz01a	15.48325	14.99291	10.15603	26.48936	3.46281
Amplaz01b	10.95376	11.12356	10.32184	21.28736	4.94210
Amplaz01c	7.00996	7.03061	10.69388	14.07653	5.31091
Amplaz01d	5.29999	5.50461	11.89401	11.25806	3.81451
Amplaz01e	7.04537	7.42424	14.65801	8.16883	5.00267
Amplaz02a	15.08129	15.64516	13.11613	26.20000	6.66573
Amplaz02b	14.92311	15.45385	11.47692	23.05385	5.22859
Amplaz02c	15.03351	14.70902	11.27869	26.38525	4.43381
Amplaz02d	13.15086	14.07273	11.22727	24.90000	5.00786
Amplaz02e	15.16017	15.86919	10.41279	25.13372	5.25124
FKH01a	13.35543	14.94231	21.15385	2.53846	2.20655
FKH01b	10.66075	11.99438	17.28090	2.08989	2.44937
FKH01c	12.25765	13.41566	19.67470	2.48193	1.68977
FKH01d	7.10203	7.45000	14.02222	3.54444	1.94118
FKH01e	9.41813	10.32192	16.34247	3.61644	3.20582
FKH02a	6.32638	6.80769	13.87179	8.61538	3.21923
FKH02b	7.53180	8.45968	14.91935	7.66129	2.52174
FKH02c	6.32782	6.85632	13.12644	11.70115	2.93551
FKH02d	5.81814	5.91096	11.47945	8.21918	2.54022
FKH02e	4.66708	4.75294	6.29412	10.07059	3.01141
Best Result	4.66708	4.75294	6.29412	2.08989	1.68977

TABLE 5. DeaSpeedDataset Detail Information about number of vehicles for each direction.

Locations	Ground Truth	
	Left	Right
FKH01a	53	25
FKH01b	57	32
FKH01c	58	25
FKH01d	45	45
FKH01e	36	37
FKH02a	28	11
FKH02b	34	28
FKH02c	35	52
FKH02d	24	49
FKH02e	45	40

D. COMPARISON ON VEHICLE DIRECTIONS

A subsequent experimental study discussed the effect of different vehicle directions on the predicted results. Table 5 lists the number of vehicles in each setup in various directions. In Indonesia, the left-side road is a vehicle that moves away from the camera, whereas the right-side road is a vehicle that moves toward the camera. The number of vehicles can affect the regression model because the machine-learning-based approach depends on the number of instances of data. In most cases, more data can improve the accuracy of predictions. However, some studies have addressed the overfitting problem. In addition, small amounts of data can cause issues such as bias or outliers, which affect accuracy. In this direction experimental study, the location uses only the second location

TABLE 6. RMSE of left-side vehicles on DeaSpeedDataset.

Locations	Average	Median	Min	Max	MLR
FKH01a	14.63181	16.60828	22.34633	2.91709	2.50059
FKH01b	12.06816	13.86384	19.04749	2.15618	1.72874
FKH01c	13.18597	14.71145	20.66273	3.00574	2.42113
FKH01d	8.74612	9.71539	16.37885	4.00555	3.03356
FKH01e	11.52654	13.08731	18.46468	6.19139	8.03970
FKH02a	6.13544	6.83674	16.44797	8.24838	3.13283
FKH02b	8.71825	9.40979	17.64686	5.01762	2.87192
FKH02c	6.28640	6.65582	16.22432	6.91169	2.40687
FKH02d	7.57455	8.11955	17.22280	7.64035	2.79337
FKH02e	5.08108	4.77202	7.43565	12.09499	4.20519
Best Result	5.08108	4.77202	7.43565	2.15618	1.72874

TABLE 7. RMSE of right-side vehicles on DeaSpeedDataset.

Locations	Average	Median	Min	Max	MLR
FKH01a	12.94517	14.32760	20.00200	6.82642	7.08019
FKH01b	10.93253	11.93210	16.31142	5.38807	2.67348
FKH01c	12.70062	13.69233	18.75100	5.00400	4.29753
FKH01d	7.23023	7.67282	13.52036	7.26483	2.18706
FKH01e	9.61626	10.16918	16.33683	4.67310	3.74763
FKH02a	12.07725	12.76537	11.28152	19.39775	38.39618
FKH02b	9.32479	11.22935	14.34648	18.49228	3.80176
FKH02c	9.83465	10.89747	12.65899	20.88706	2.45162
FKH02d	6.59483	6.88980	10.46179	12.94178	2.59403
FKH02e	7.23574	7.60222	7.38072	16.40503	4.01973
Best Result	6.59483	6.88980	7.38072	5.00400	2.18706

because, in this location, both directions can be recorded simultaneously in the same video frame.

Based on Tables 6 and 7, the best results for the left-side vehicles were more dominant than those for the right-side vehicles. All the RMSE values in the best results of left-side vehicles are smaller than those of right-side vehicles, except for the minimum speed with different values of approximately 0.05493 km/h, which means that the installation of the camera for recording the moving vehicle away has better results than moving toward. The best RMSE for the left-side vehicles was 1.72874 km/h, as determined using the MLR method. In the right-side RMSE table, the largest RMSE in the MLR speed is the *FKH02a* setup, which is approximately 38.39618 km/h. A possible reason for this is that the amount of training data is too small. Table 5 shows that the number of vehicles was only 11, which means that the training data was only 70% of the 11 instances.

E. STATE-OF-THE-ART SPEED ESTIMATION STUDY

The last experiment compared the proposed study with previous speed estimation research. Table 8 compares the results of previous and proposed studies on speed estimation. The table contains the author, year of published research, dataset used, and mean error results. Some studies used MAE, whereas others used RMSE. The unit is expressed in kilometers per

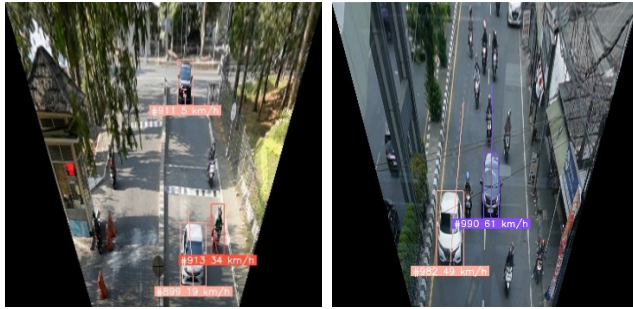


FIGURE 11. (Left) Detected motorcycle, (Right) miss vehicle classification.

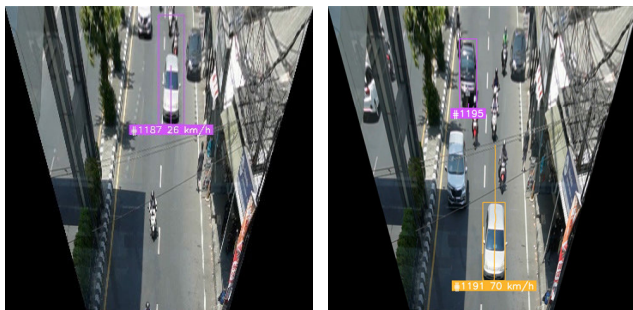


FIGURE 12. (Left) White car in id #1187, (Right) the same car but change into id #1191.

hour (km/h). The detailed methods of state-of-the-art speed estimation research are discussed in Section II, which examines related works. In Table 8, it can be observed that the proposed method outperforms other state-of-the-art methods, except for OptCalib (2019) [17]. The best result of the proposed method was 1.68977 km/h, whereas that of OptCalib (2019) [17] was 1.43000 km/h, with a difference only 0.25977 km/h. In addition, the comparison results of [16] was the results implemented by [17] in BrnoCompSpeed Dataset using the same speed measurement method.

F. SOME COMMON MISTAKES

For evaluation purposes, the authors assumed that only four-wheeled vehicles, including cars, trucks, and buses, could be evaluated. The detection is not limited to specific classes; only the person class is limited, so motorcycles are still detected even though they are not included in our evaluation. The ground truth is only considered in four-wheeled vehicles because limited human power can only handle large vehicles. Furthermore, the lousy conditions of a video affected by weather conditions, such as high light intensity, can cause classes of vehicles to be misidentified. However, because the research is not concerned with classification tasks, the evaluation is still applied as long as the detection and predicted speed are working. The details are presented in Figure 11.

Furthermore, light conditions can affect the model performance. At the *Amplaz* location, the bridge view was covered by glass; because the glass was clear, there was no crucial problem. However, in the *Amplaz02* setup, when the

TABLE 8. Comparison between state-of-the-art speed estimation research and the proposed method.

Author	Dataset	RMSE	MAE
Nurhad et al. 2013 [22]	Primer	-	7.63000
Sina et al. 2013 [23]	Primer	-	3.32500
Kumar et al. 2014 [24]	Primer	-	2.00000
Dubská et al. 2014 [16]	BrnoCompSpeed	-	8.59000
Lin et al. 2015 [28]	Primer	-	3.00000
Filipiak et al. 2016 [25]	Primer	-	2.00000
Kumar et al. 2018 [18]	AIC18	15.35314	-
Tang et al. 2018 [19]	AIC18	6.59235	-
Huang et al. 2018 [20]	AIC18	13.85468	-
Hua et al. 2018 [21]	AIC18	19.48819	-
Tran et al. 2018 [13]	AIC18	14.34634	-
Bouzia et al. 2018 [29]	Primer	-	2.00000
OptScale 2019 [17]	BrnoCompSpeed	-	1.71000
OptScaleVP2 2019 [17]	BrnoCompSpeed	-	15.66000
OptCalib 2019 [17]	BrnoCompSpeed	-	1.43000
OptCalibVP2 2019 [17]	BrnoCompSpeed	-	2.43000
Trivedi et al. 2021 [26]	Primer	-	2.87500
Wu et al. 2021 [27]	Primer	-	3.20000
Yohann et al. 2023 [6]	BrnoCompSpeed	2.56125	-
	AIC18	4.08289	-
Kamil et al. 2023 [3]	Primer	3.92600	3.15500
Proposed Method	DeaSpeedDataset	2.37897	1.68977

light intensity was high, the reflection reflected by the glass slightly interfered with the detection and tracking results. Figure 12 shows the same car in a sequence of frames, but it was detected and tracked as a different vehicle. Because the vehicle speed system uses a consecutive approach, the speed value can still be detected, and the speed value used is closer to the ground truth. Moreover, the reflection affects the precision of the bounding box. As shown in Figure 13, the upper image illustrates the shrunken bounding box in the bus, and the lower image shows the widened bounding box in the car. The imprecision occasion can affect speed prediction when the bounding box suddenly shrinks or widens in the middle of the predicted speed sequence. Therefore, further development is needed, especially in terms of detection, so that the noise from the glass reflections does not significantly affect the results. To improve detection quality, the transfer learning method can be applied. Meanwhile, in terms of occlusion, because the video is taken from the above vehicles, the images of each vehicle do not overlap with each other; thus, occlusion infrequently occurs and does not affect model performance.

In the video result link provided, it can be observed that there are missing vehicles in successive frames. In this study, speed estimation uses consecutive frame scenarios, and some missing vehicle speed values can be recovered by the other frame. Even though the name of the method is a consecutive frame, the speed value formulation is calculated by the two points in the adjacent frames. If the vehicle is suddenly missing, the speed calculation starts again in the frame when



FIGURE 13. Unprecise bounding box, (Upper) shrink bounding box, (Lower) widened bounding box.

the vehicle is re-detected. Missing detection and tracking occurred because the video used for the detection application was the result of rectification by the homography matrix. Therefore, the shape of the vehicle also changed.

V. CONCLUSION

In conclusion, this research proposes a vehicle speed estimation method, which is followed by a deep image homography transformation network for transforming monocular video images into a bird's-eye view, object detection using YOLOv8x, which outperformed the other YOLOv8 versions; tracking using ByteTrack; and estimating vehicle speed in consecutive frames using statistical approaches: average, median, minimum, and maximum. The statistical methods were then improved using Multiple Linear Regression. Furthermore, the best RMSE was approximately 2.37897 km/h, and MAE was approximately 1.68977 km/h. These results were compared with those of state-of-the-art studies and outperformed the results. In future research, because the pixel per meter value is generated manually in this research, it can be improved by detecting the value automatically. Moreover,

standard Automatic Number Plate Recognition (ANPR) camera installation will be possible in future work because it is an important application and useful for speed estimation systems.

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interests include digital image processing, pattern recognition, machine learning, and computer vision.



Sciences, Universitas Gadjah Mada. His research interests include digital image processing, pattern recognition, machine learning, computer vision, surveillance systems, and software engineering.



recognition, instrumentation, and sensor networks.



ent intelligence. He was the Director or an AdCom Member of the Institute of Control, Robotics and Systems and the Society of Instrument and Control Engineers, the IEEE IES Technical Committee on Human Factors Chair, an AdCom Member, and the Secretary, until 2019.

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